

Supporting Information

Molar Mass Determination for Small and Large Molecules using Diffusion-Ordered Spectroscopy

Wolf Hiller,* Bastian Grabe and Jan Schonert

TU Dortmund University
Faculty of Chemistry and Chemical Biology
Otto-Hahn-Str. 4a, 44227 Dortmund

*Corresponding Author:

Wolf Hiller - TU Dortmund University, Faculty of Chemistry and Chemical Biology, 44227 Dortmund, Germany

Email:

Content:

S2: Molar masses of the polymers
S3: Molar masses of small molecules
S4-S6: Experimental conditions of DOSY measurements
S6: Calculated parameters with DOSY and dynamic viscosities
S7: Temperature dependences of dynamic viscosities
S8: References
S9-S14: Diffusion coefficients in dependence of molar masses
S15: Experimental vs. calculated diffusion coefficients

1) Tables:

Table S1. Molar masses (in g/mol) of PS-, PMMA, PI, P2VP and PEO homopolymers used for the DOSY calibration.

Polymer	Sample	M _p	M _w	M _n	PDI
PS	ps1	1306	1220	1120	1.09
	ps2	3500	3510	3350	1.05
	ps3	8680	8670	8420	1.03
	ps4	19700	19100	18200	1.05
	ps5	34800	34000	32700	1.04
	ps6	66000	62500	59300	1.05
	ps7	130000	125000	120000	1.04
	ps8	304000	294000	282000	1.04
	ps9	552000	554000	532000	1.04
	ps10	1240000	1170000	1100000	1.06
PMMA	pmma1	2160	2130	1920	1.11
	pmma2	5980	5930	5590	1.06
	pmma3	12900	12800	12300	1.04
	pmma4	22800	21200	18700	1.13
	pmma5	41400	40300	38100	1.06
	pmma6	88500	86700	83700	1.04
	pmma7	202000	187000	175000	1.07
	pmma8	520000	517000	501000	1.03
	pmma9	988000	936000	850000	1.10
PI	pi1	1030	1070	944	1.13
	pi2	2200	2190	2030	1.08
	pi3	4620	4610	4450	1.04
	pi4	12100	12000	11800	1.02
	pi5	23500	23600	23300	1.01
	pi6	42300	42500	42000	1.01
	pi7	111000	111000	110000	1.01
	pi8	226000	231000	227000	1.02
	pi9	443000	443000	429000	1.03
	pi10	1180000	1090000	1010000	1.08
P2VP	p2vp1	1020	1050	867	1.21
	p2vp2	2450	2490	2180	1.14
	p2vp3	5780	5790	4630	1.25
	p2vp4	11500	12200	11500	1.06
	p2vp5	19600	20500	19600	1.04
	p2vp6	42800	43100	40500	1.06
	p2vp7	118000	120000	113000	1.06
	p2vp8	243000	249000	232000	1.07
	p2vp9	539000	535000	488000	1.10
	p2vp10	1160000	1200000	1040000	1.15
PEO	peo1	1030	1020	978	1.04

peo2	3450	3450	3340	1.03
peo3	6530	6200	5860	1.06
peo4	12600	11400	10300	1.11
peo5	26100	25800	22100	1.17
peo6	42700	40100	30700	1.31
peo7	99000	103000	91800	1.12
peo8	217000	220000	196000	1.12
peo9	504000	504000	461000	1.09
peo10	1180000	1200000	1110000	1.08

Table S2. Molar masses (in g/mol) of small molecules used for the DOSY calibration. The samples are separated according to their shapes: CS=compact spheres, DSE=dissipated spheres and ellipsoids, ED=expanded discs (nomenclature according to ref (1,2)).

Shape	Sample	M [g/mol]
CS	Cyclopentane	70
	THF	72
	TMS	88
	Methyl- <i>tert</i> -butylether (MTB)	88
	Tetramethylbutane (TMB)	114
	Adamantane	136
	N(SiMe ₃) ₃	234
	Si(SiMe ₃) ₄	321
	Toluene	92
	Diisopropylether	102
DSE	Indene	116
	Naphthalene	128
	1,3-Indandione	146
	2-Phenylpyridine	155
	Tetramethoxypropane	164
	Diphenylacetylene	178
	Diphenylsulfoxide	202
	1-Phenylnaphthalene	204
	Triphenylmethane	244
	Tri-(<i>o</i> -tolyl)-phosphine	304
ED	Hexaphenyltrisiloxane	595
	BINAP	623
	Anthracene	178
	Acridine	179
	9-Methylantracene	192
	Pyrene	202
	Anthrachinone	208
	Triphenylene	228
	3,4-Dihydrobenzophenanthrene	281
	Tetraphenylnaphthalene	433

Table S3. Pulsed field gradient conditions for all polymers and solvents: Gradient pulse length given as little delta (LD) and diffusion delay as big delta (BD) for the convection compensated pulse sequence dstebpgp3s

Sample	Gradients	CDCl ₃	C ₆ D ₆	CD ₂ Cl ₂	THF	Toluene	Acetone
ps1	LD/ms	2.0	2.0	1.6	2.0	2.0	1.6
	BD/ms	76.73	76.73	77.27	76.73	76.73	57.27
ps2	LD/ms	2.0	2.0	1.6	2.0	2.0	1.6
	BD/ms	76.73	76.73	77.27	76.73	76.73	57.27
ps3	LD/ms	2.6	2.6	2.4	2.6	2.6	2.0
	BD/ms	95.93	95.93	76.2	95.93	95.93	76.73
ps4	LD/ms	3.2	3.2	3.0	3.2	3.2	2.6
	BD/ms	95.13	95.13	95.4	95.13	95.13	75.93
ps5	LD/ms	3.8	3.8	3.6	3.8	3.8	3.0
	BD/ms	94.33	94.33	94.6	94.33	94.33	75.4
ps6	LD/ms	4.2	4.2	4.0	4.2	4.2	-
	BD/ms	113.8	113.8	94.07	113.8	113.8	-
ps7	LD/ms	5.6	5.6	5.6	5.6	-	-
	BD/ms	111.93	91.93	141.93	91.93	-	-
ps8	LD/ms	5.6	5.6	5.6	5.6	-	-
	BD/ms	191.93	191.93	251.93	191.93	-	-
ps9	LD/ms	5.6	5.6	5.6	5.6	-	-
	BD/ms	291.93	241.93	251.93	241.93	-	-
ps10	LD/ms	5.6	5.6	5.6	5.6	-	-
	BD/ms	291.93	391.93	341.93	391.93	-	-
Sample	Gradients	CDCl ₃	C ₆ D ₆	CD ₂ Cl ₂	THF	DMSO	-
pmma1	LD/ms	2.0	2.0	1.6	2.0	3.6	-
	BD/ms	76.73	76.73	77.27	76.73	74.6	-
pmma2	LD/ms	2.0	2.0	2.0	2.2	3.8	-
	BD/ms	76.73	76.73	76.73	76.47	94.33	-
pmma3	LD/ms	2.6	2.6	2.4	2.6	4.0	-
	BD/ms	95.93	95.93	96.2	95.93	94.07	-
pmma4	LD/ms	3.2	3.2	3.0	3.2	4.6	-
	BD/ms	95.13	95.13	95.4	95.13	113.27	-
pmma5	LD/ms	3.8	3.8	3.6	3.8	5.2	-
	BD/ms	94.33	94.33	94.6	94.33	122.47	-
pmma6	LD/ms	4.2	4.2	4.0	4.2	5.6	-
	BD/ms	113.8	113.8	94.07	113.8	141.93	-
pmma7	LD/ms	5.6	5.6	5.6	5.6	-	-
	BD/ms	141.93	141.93	141.93	91.93	-	-
pmma8	LD/ms	5.6	5.6	5.6	5.6	-	-
	BD/ms	291.93	291.93	251.93	191.93	-	-
pmma9	LD/ms	5.6	5.6	5.6	5.6	-	-
	BD/ms	391.93	391.93	341.93	341.93	-	-
Sample	Gradients	CDCl ₃	C ₆ D ₆	CD ₂ Cl ₂	THF	-	-
pi1	LD/ms	1.6	2.0	1.6	1.8	-	-
	BD/ms	77.27	76.73	77.27	77.0	-	-
pi2	LD/ms	2.0	2.0	1.8	2.0	-	-

	BD/ms	77.0	76.73	77.0	76.73	-	-
	LD/ms	2.4	2.4	2.4	2.6	-	-
pi3	BD/ms	96.2	96.2	96.2	95.93	-	-
	LD/ms	3.0	3.2	2.6	3.2	-	-
pi4	BD/ms	95.4	95.13	95.93	95.13	-	-
	LD/ms	3.6	4.0	3.6	3.8	-	-
pi5	BD/ms	94.6	94.07	94.6	94.33	-	-
	LD/ms	4.2	4.4	4.2	4.2	-	-
pi6	BD/ms	113.8	113.53	113.8	113.8	-	-
	LD/ms	4.6	5.2	4.6	4.8	-	-
pi7	BD/ms	143.27	142.47	113.27	143.0	-	-
	LD/ms	5.6	5.6	-	5.6	-	-
pi8	BD/ms	191.93	191.93	-	161.93	-	-
	LD/ms	5.6	5.6	-	5.6	-	-
pi9	BD/ms	291.93	291.93	-	261.93	-	-
	LD/ms	5.6	5.6	-	5.6	-	-
pi10	BD/ms	491.93	491.93	-	441.93	-	-
Sample	Gradients	C₆D₆	CD₂Cl₂	THF	MeOD	-	-
	LD/ms	1.8	1.6	1.6	1.8	-	-
p2vp1	BD/ms	77.0	77.27	77.27	77.0	-	-
	LD/ms	2.0	1.6	1.8	2.0	-	-
p2vp2	BD/ms	76.73	77.27	77.0	76.73	-	-
	LD/ms	2.4	2.0	2.0	2.4	-	-
p2vp3	BD/ms	96.2	96.73	96.73	96.2	-	-
	LD/ms	2.8	2.4	2.4	2.8	-	-
p2vp4	BD/ms	95.67	96.2	96.2	95.67	-	-
	LD/ms	3.4	2.8	2.8	3.2	-	-
p2vp5	BD/ms	94.87	105.67	105.67	105.13	-	-
	LD/ms	4.0	3.4	3.4	3.8	-	-
p2vp6	BD/ms	114.07	114.87	114.87	114.33333	-	-
	LD/ms	4.6	4.0	4.0	4.4	-	-
p2vp7	BD/ms	123.27	124.07	124.07	123.53	-	-
	LD/ms	5.6	5.6	5.6	-	-	-
p2vp8	BD/ms	141.93	141.93	141.93	-	-	-
	LD/ms	5.6	5.6	5.6	-	-	-
p2vp9	BD/ms	191.93	191.93	191.93	-	-	-
	LD/ms	5.6	5.6	5.6	-	-	-
p2vp10	BD/ms	341.93	341.93	341.93	-	-	-
Sample	Gradients	MeOD	D₂O	CD₃CN	THF	Acetone	-
	LD/ms	2.0	2.6	1.6	2.0	1.6	-
peo1	BD/ms	76.73	75.93	77.27	76.73	57.27	-
	LD/ms	2.0	2.6	1.6	2.0	1.6	-
peo2	BD/ms	76.73	75.93	77.27	76.73	57.27	-
	LD/ms	2.6	3.2	2.4	2.4	2.0	-
peo3	BD/ms	95.93	95.13	76.2	96.2	76.73	-
peo4	LD/ms	3.2	3.8	3.0	3.0	2.6	-

peo5	BD/ms	95.13	94.33	95.4	95.4	75.93	-
	LD/ms	3.8	4.6	3.6	3.6	3.0	-
peo6	BD/ms	94.33	93.27	94.6	94.6	75.4	-
	LD/ms	4.2	5.2	4.0	4.0	-	-
peo7	BD/ms	113.8	112.47	94.07	114.07	-	-
	LD/ms	5.6	5.6	5.6	5.6	-	-
peo8	BD/ms	141.93	241.93	91.93	141.93	-	-
	LD/ms	5.6	5.6	5.6	5.6	-	-
peo9	BD/ms	291.93	341.93	161.93	191.93	-	-
	LD/ms	5.6	5.6	5.6	5.6	-	-
peo10	BD/ms	291.93	491.93	241.93	291.93	-	-
	LD/ms	5.6	5.6	5.6	5.6	-	-
	BD/ms	491.93	641.93	441.93	491.93	-	-

Table S4a. Parameters b (Flory parameter) and c of the power scaling $D=c \cdot M^{-b}$ determined via DOSY and the dynamic viscosities (in Pa·s) of the solvents for small molecules of different shapes at T=298 K: compact spheres (CS), dissipated spheres and ellipsoids (DSE) and expanded discs (ED) (parameters b and c are calculated from diffusion data of ref (1) reproduced with permission of the Royal Society of Chemistry and ref (2) reproduced with permission of John Wiley and Sons, respectively). The dynamic viscosities are related to the references and temperature dependences of Table S5.

Sample	THF-d8	C ₆ D ₆	CDCl ₃	CD ₂ Cl ₂	Toluene-d ₈	DMSO-d ₆	C ₆ D ₁₂
CS / b	0.50087	0.46669	0.4635	0.4837	0.5065	0.61984	0.58898
CS / c	1.8988E-8	1.562E-8	1.5319E-8	2.1503E-8	1.7992E-8	1.0117E-8	1.9022E-8
CS / η	0.0005011	0.0006381	0.0005444	0.0004137	0.000619	0.002185	0.0009597
DSE / b	0.61287	0.62975	0.57938	0.56926	0.61902	0.72055	0.7025
DSE / c	3.4413E-8	3.5605E-8	2.6851E-8	3.3222E-8	3.1356E-8	1.7684E-8	3.8083E-8
DSE / η	0.0005011	0.0006381	0.0005444	0.0004137	0.000619	0.002185	0.0009597
ED / b	0.78013	0.76497	0.61563	0.63314	0.8235	0.89594	1.0642
ED / c	8.8757E-8	7.6004E-8	3.5927E-8	5.1783E-8	9.8044E-8	4.6005E-8	2.8854E-7
ED / η	0.0005011	0.0006381	0.0005444	0.0004137	0.000619	0.002185	0.0009597

Table S4b. Parameters b (Flory parameter) and c of the power scaling $D=c \cdot M^{-b}$ determined via DOSY and the dynamic viscosities (in Pa·s) of the solvents for PS, PMMA, PI, P2VP and PEO at T=298.15 K. The dynamic viscosities are related to the references and temperature dependences of Table S5.

Sample	THF-d8	C ₆ D ₆	CDCl ₃	CD ₂ Cl ₂	Toluene-d ₈	Acetone-d ₆
PS / b	0.5604	0.56041	0.56239	0.56162	0.53106	0.45209
PS / c	3.1796E-8	2.5445E-8	2.9966E-8	3.92E-8	2.1376E-8	2.0934E-8
PS / η	0.000501	0.000636	0.000544	0.000413	0.0006179	0.000326
	THF-d8	C ₆ D ₆	CDCl ₃	CD ₂ Cl ₂	DMSO-d ₆	-
PMMA/b	0.5624	0.56913	0.58372	0.58736	0.44424	-
PMMA/c	3.6793E-8	2.9624E-8	3.8316E-8	5.1798E-8	3.1205E-9	-
PMMA/ η	0.000501	0.000636	0.000544	0.000413	0.00218	-
	THF-d8	C ₆ D ₆	CDCl ₃	CD ₂ Cl ₂	-	-
PI / b	0.60811	0.61614	0.62721	0.57317	-	-
PI / c	4.672E-8	4.0871E-8	5.0317E-8	4.1675E-8	-	-
PI / η	0.000501	0.000636	0.000544	0.000413	-	-
	THF-d8	C ₆ D ₆	CD ₂ Cl ₂	MeOD	-	-
P2VP/b	0.5049	0.48883	0.53396	0.4789	-	-
P2VP/c	1.8199E-8	1.2484E-8	2.6449E-8	1.1295E-8	-	-

P2VP/η	0.000501	0.000636	0.000413	0.000602	-	-
	THF-d8	CD₃CN	D₂O	MeOD	Acetone-d6	-
PEO / b	0.58062	0.59613	0.5723	0.57021	0.47448	-
PEO / c	3.3196E-8	5.225E-8	1.3268E-8	2.5609E-8	2.1208E-8	-
PEO / η	0.000501	0.000357	0.001098	0.000602	0.000326	-

Table S4c. Parameters b (Flory parameter) and c of the power scaling $D=c \cdot M^{-b}$ determined via DOSY and the dynamic viscosities (in Pa·s) of the solvents for PMMA in C₆D₆ and PEO in D₂O at variable temperatures in Kelvin. The dynamic viscosities are related to the references and temperature dependences of Table S5.

Sample	298.15 K	305.15 K	312.15 K	319.15 K	326.15 K	333.15 K
PMMA/b	0.52734	0.52996	0.52552	0.53221	0.51415	0.51461
PMMA/c	2.0111E-8	2.3805E-8	2.58E-8	3.0718E-8	2.9217E-8	3.2651E-8
PMMA/η	0.000636	0.0005771	0.0005253	0.0004801	0.0004405	0.0004057
Sample	298.15 K	303.15 K	313.15 K	323.15 K	333.15 K	-
PEO / b	0.5723	0.55987	0.56339	0.55747	0.55169	-
PEO / c	1.3268E-8	1.4378E-8	1.9152E-8	2.3712E-8	2.7776E-8	-
PEO / η	0.001098	0.0009738	0.0008001	0.0006653	0.0005595	-

Table S5. Temperature dependences of the dynamic viscosities in Pa·s calculated with $\log(\eta) = \frac{d}{T} + e$.

Solvent	d	e	literature
THF-d8	404.06770	-4.65605	ref. (3)
C ₆ D ₆	555.68004	-5.05978	ref. (3)
CDCl ₃	381.31056	-4.54363	ref. (4)
CD ₂ Cl ₂	428.82173	-4.82233	ref. (5)
Toluene-d8	442.06333	-4.69177	ref. (6,7)
DMSO-d6	756.47082	-5.19904	ref. (3)
C ₆ D ₁₂	653.21399	-5.20988	ref. (8)
MeOD	549.48078	-5.06149	ref. (5)
D ₂ O	810.387038	-5.68473	ref. (9,10)
CD ₃ CN	388.56288	-4.74954	ref. (3)
Acetone-d6	376.24429	-4.749	ref. (3)

References:

- (1) Bachmann, S.; Neufeld, R.; Dzemski, M.; Stalke, D. New External Calibration Curves (ECCs) for the Estimation of Molecular Weights in Various Common NMR Solvents. *Chemistry – A European Journal* **2016**, 22, 8462–8465.
- (2) Neufeld, R.; Stalke, D. Accurate Molecular Weight Determination of Small Molecules via DOSY-NMR by Using External Calibration Curves with Normalized Diffusion Coefficients. *Chem Sci* **2015**, 6, 3354–3364.
- (3) Holz, M.; Mao, Xi-an; Seiferling, D.; Sacco, A. Experimental study of dynamic isotope effects in molecular liquids: Detection of translation-rotation coupling. *J. Chem. Phys.* **1996**, 104, 669–679.
- (4) Evans, R.; Dal Poggetto, G.; Nilsson, M.; Morris, G. A. Improving the Interpretation of Small Molecule Diffusion Coefficients. *Anal. Chem.* **2018**, 90, 3987–3994.
- (5) Kennedy, K. G.; Miles, D. T. *J Undergrad Chem Res* **2004**, 4, 145–150.
- (6) Wilbur, D. J.; Jonas, J. Fourier transform NMR in liquids at high pressure. III. Spin–lattice relaxation in toluene-d8. *J. Chem. Phys.* **1975**, 62, 2800–2807.
- (7) Artaki, I.; Jonas, J. Pressure effect on the coupling between rotational and translational motions of supercooled viscous fluids. *J. Chem. Phys.* **1985**, 82, 3360–3370.
- (8) Dixon, J. A.; Schiessler, R. W. Viscosities of Benzene-d6 and Cyclohexane-d12. *J. Phys. Chem.* **1954**, 58, 430–432.
- (9) Hardy, R. C.; Cottington, R. L. *J. Research National Bureau of Standards* **1949**, 42, 573-578.
- (10) Harris, K. R.; Woolf, L. A. Temperature and Volume Dependence of the Viscosity of Water and Heavy Water at Low Temperatures. *J. Chem. Eng. Data* **2004**, 49, 1064-1069.

2) Figures:

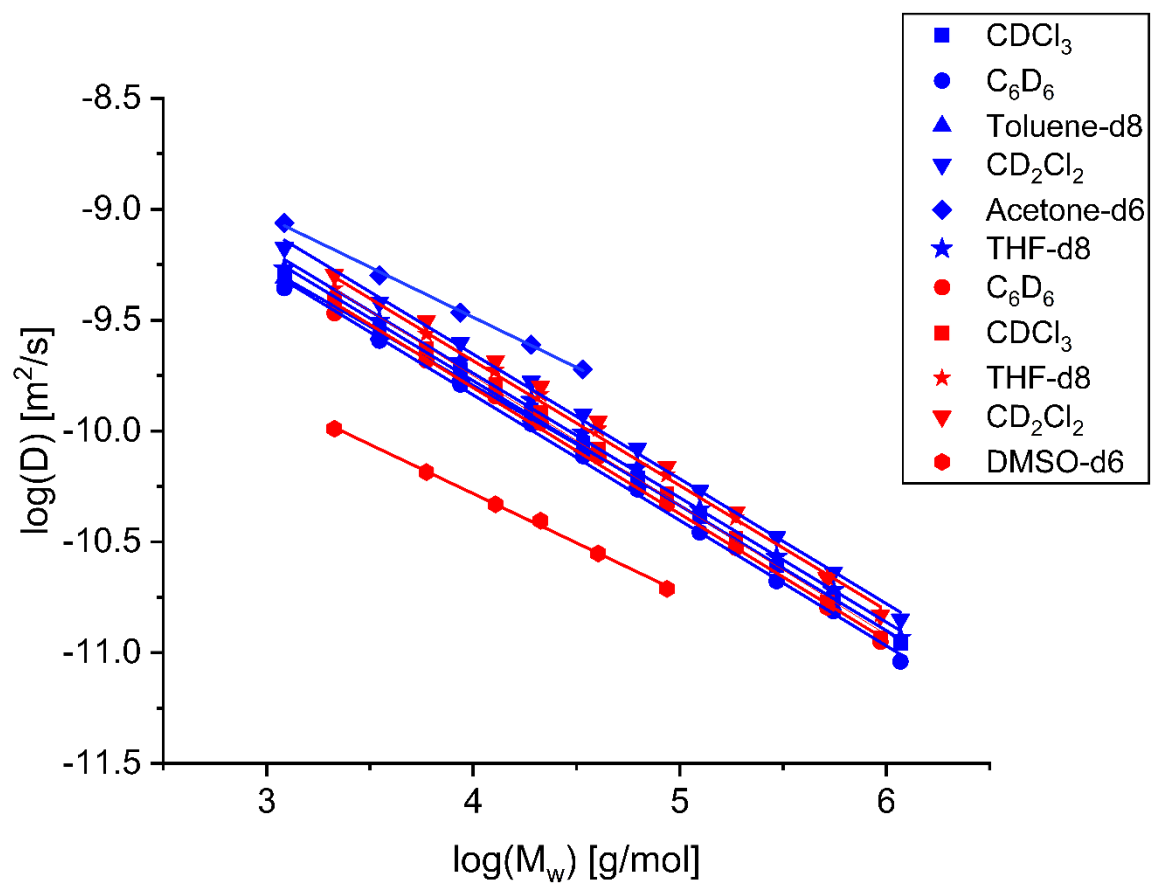


Figure S1. Diffusion coefficients of PS (blue symbols) and PMMA (red symbols) in dependence of the molar mass for different solvents.

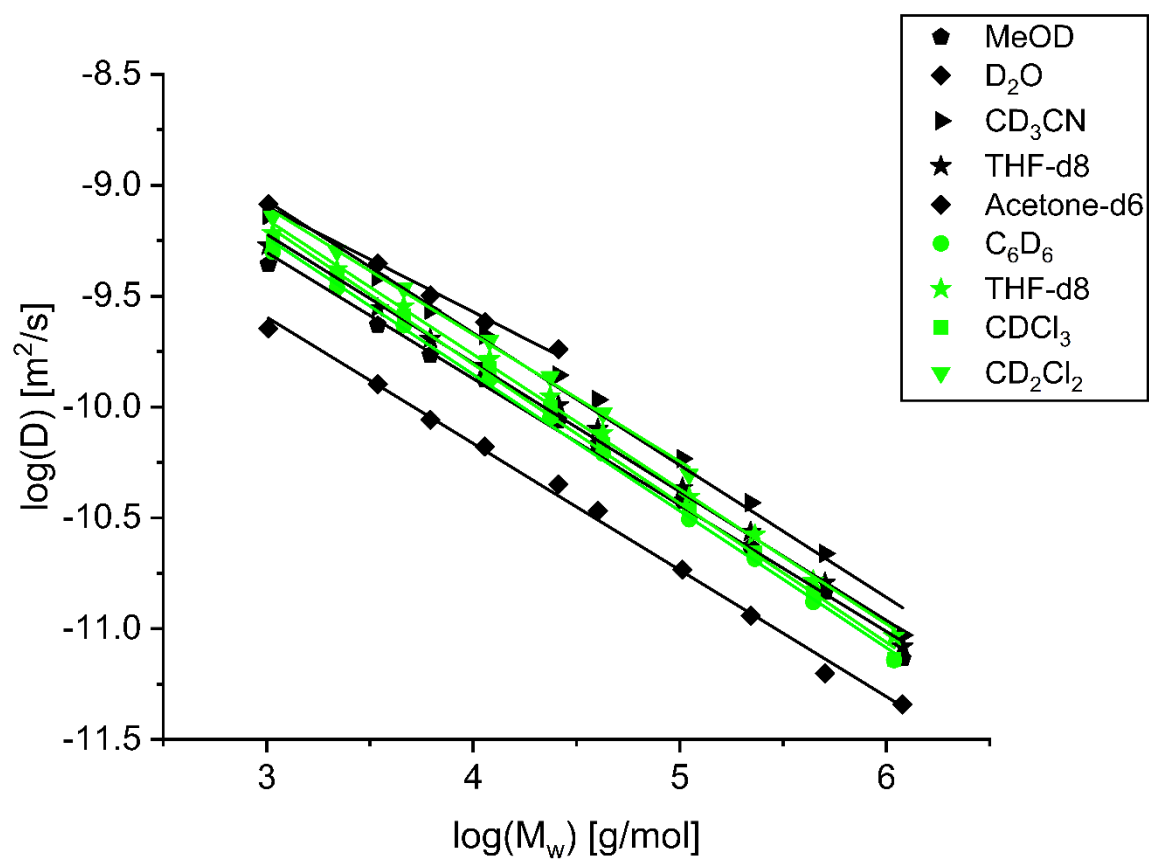


Figure S2. Diffusion coefficients of PEO (black symbols) and PI (green symbols) in dependence of the molar mass for different solvents.

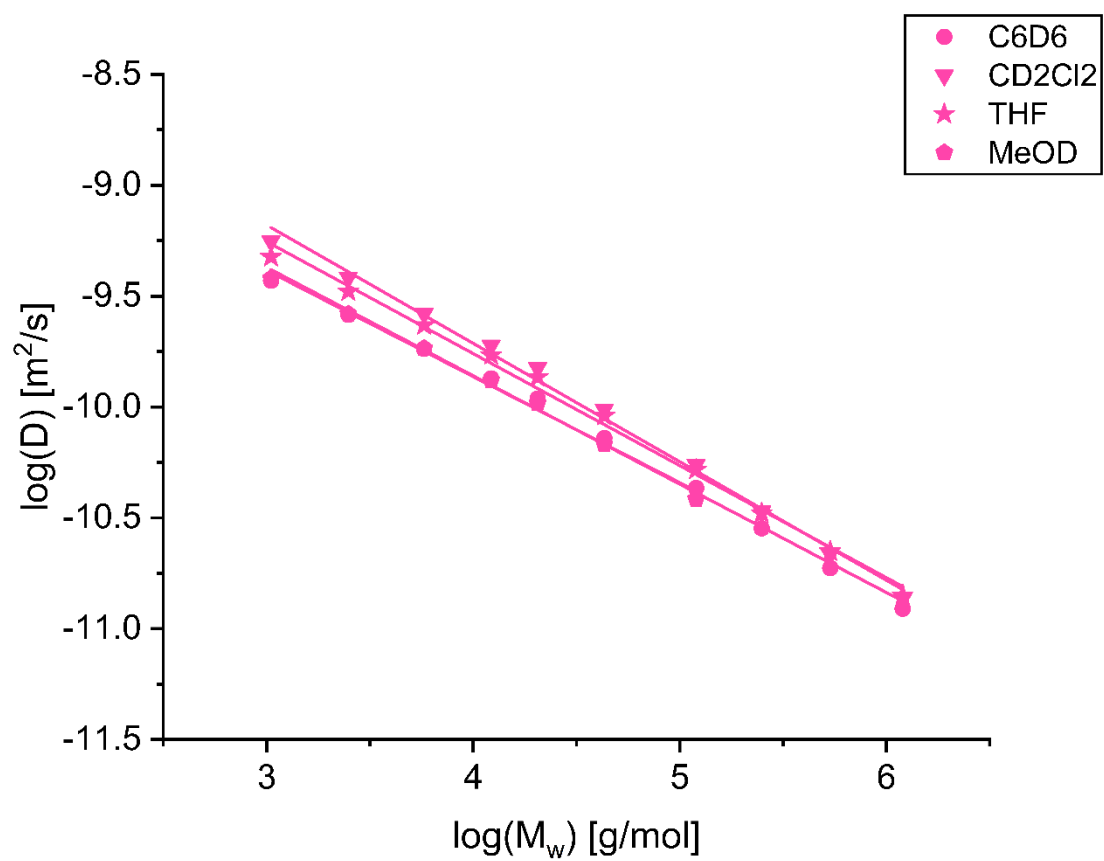


Figure S3. Diffusion coefficients of P2VP in dependence of the molar mass for different solvents.

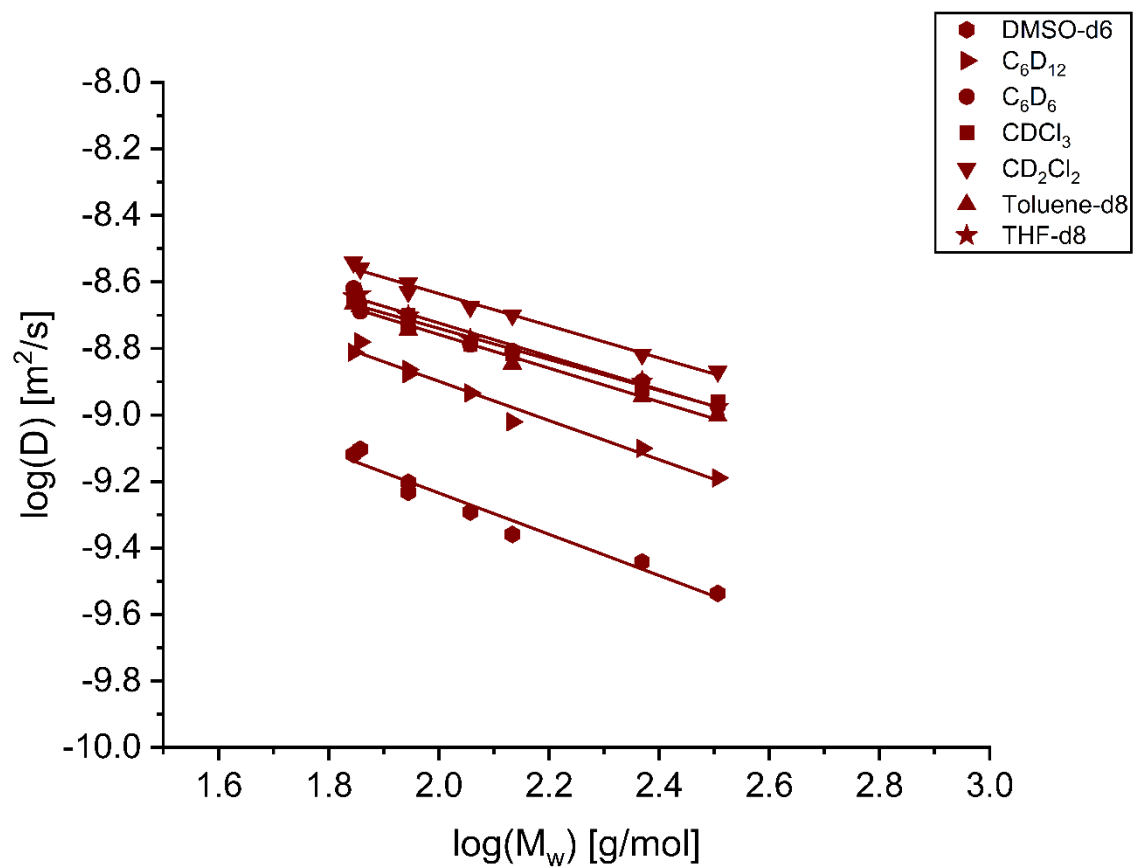


Figure S4: Diffusion coefficients of small molecules (compact spheres) in dependence of the molar mass for different solvents (reproduced from ref (1) with permission of the Royal Society of Chemistry and reproduced from ref (2) with permission of John Wiley and Sons).

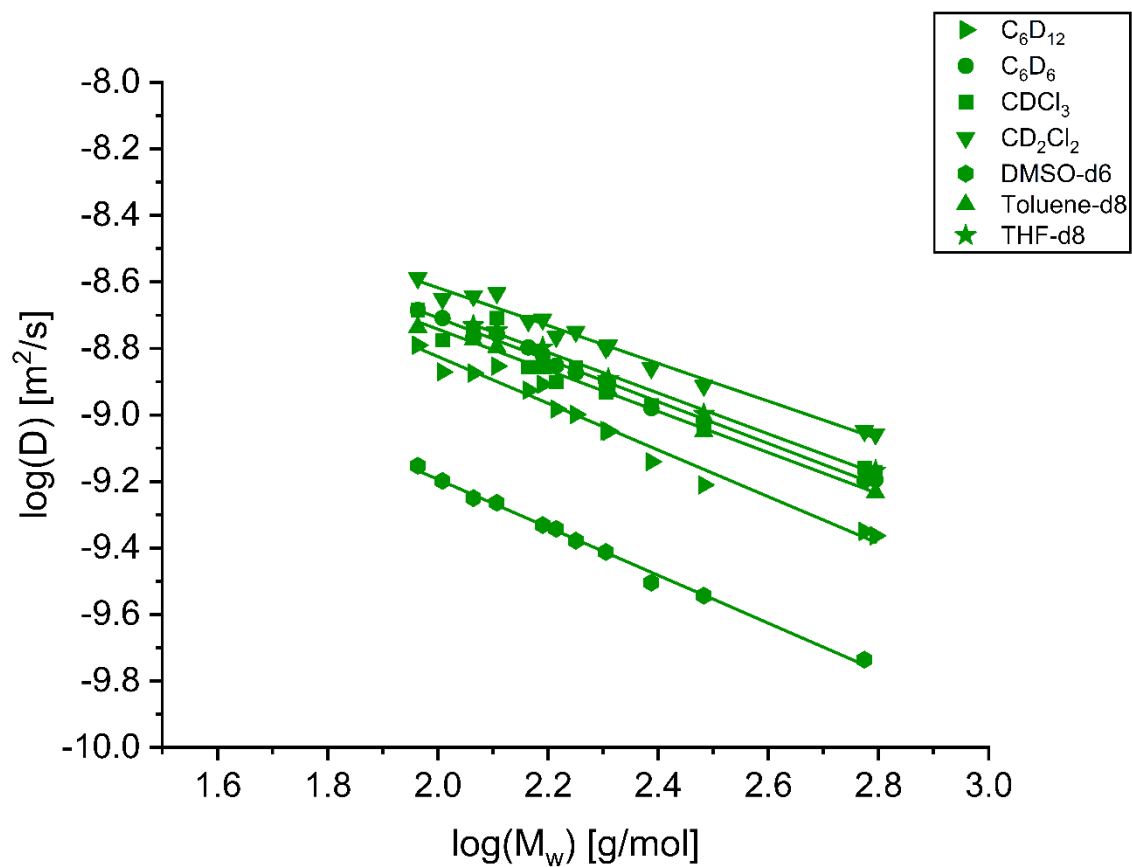


Figure S5: Diffusion coefficients of small molecules (dissipated spheres and ellipsoids) in dependence of the molar mass for different solvents (reproduced from ref (1) with permission of the Royal Society of Chemistry and reproduced from ref (2) with permission of John Wiley and Sons).

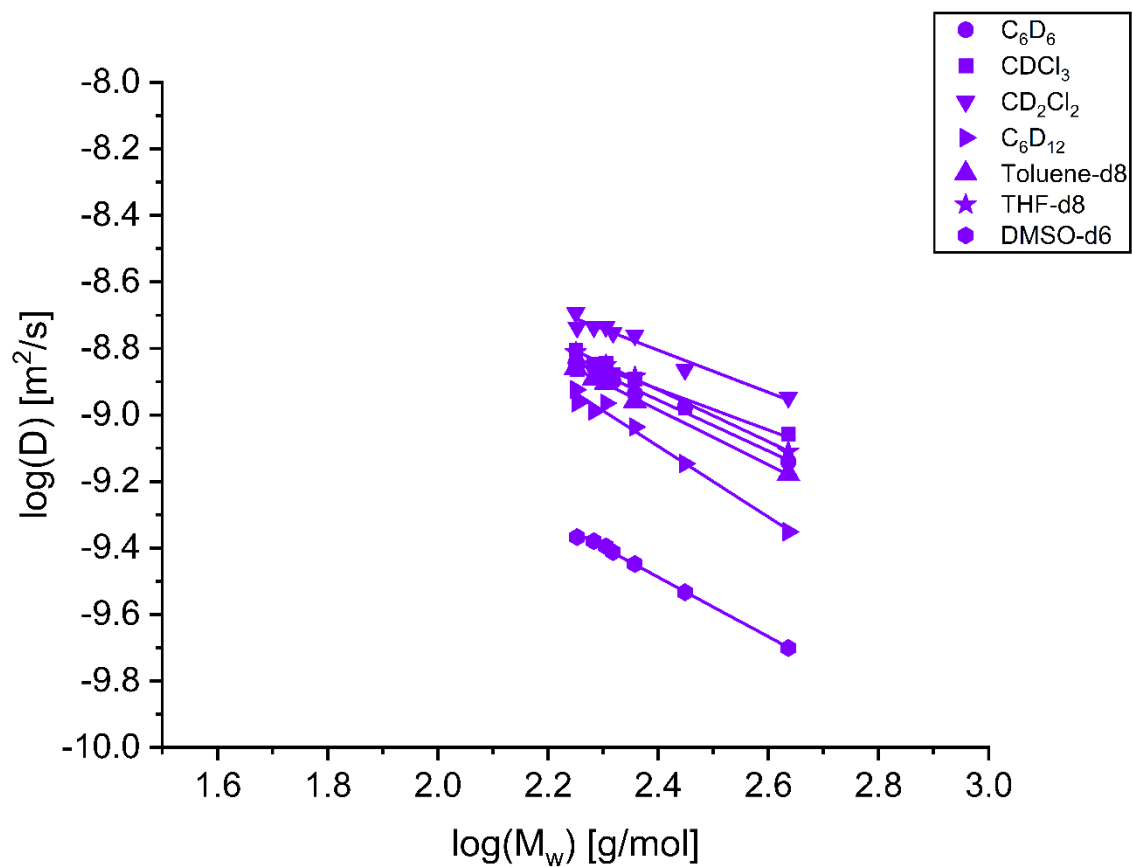


Figure S6. Diffusion coefficients of small molecules (expanded discs) in dependence of the molar mass for different solvents (reproduced from ref (1) with permission of the Royal Society of Chemistry and reproduced from ref (2) with permission of John Wiley and Sons).

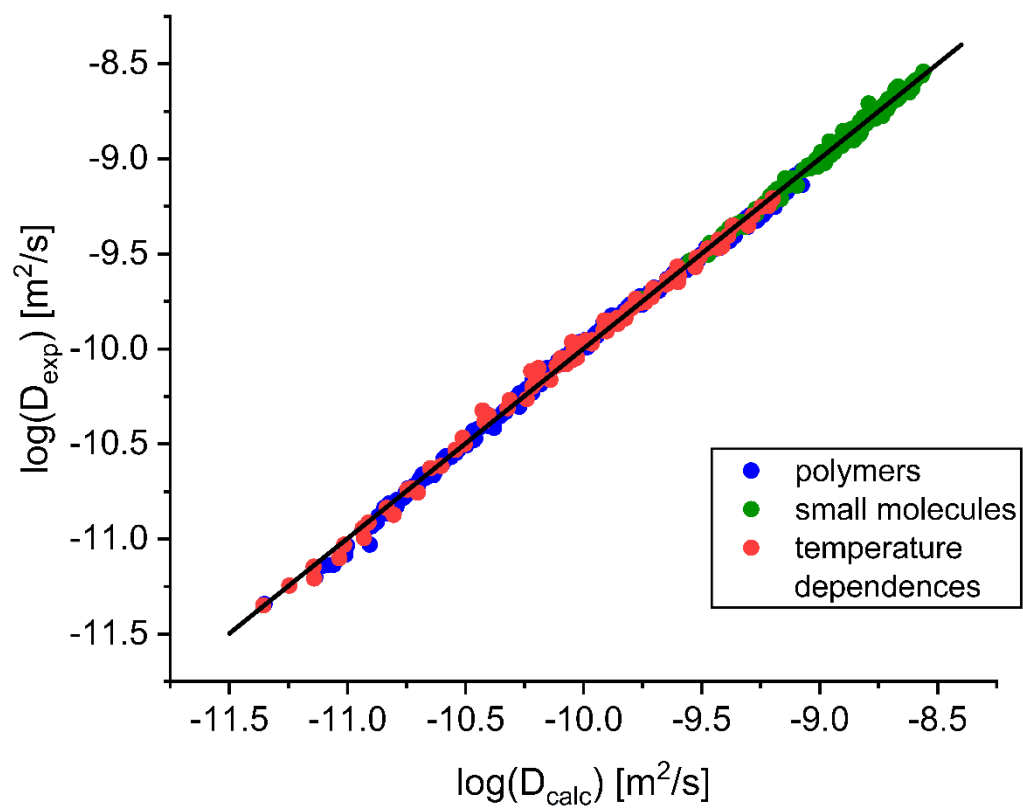


Figure S7: Experimental diffusion coefficients via equation (10) (see main text) vs. calculated diffusion coefficients for all small molecules (green circles), polymers (blue circles) and temperature dependences (red circles), respectively.